

Nearly representative Hamilton cycles with error $O(k \log k)$: an answer to Problem 6.4 of Hogan, Scott and Tsarev

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Machine-generated note

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Abstract

For a graph G , a subgraph G' and an edge labelling $h: E(G) \rightarrow \mathbb{R}^k$ with $\|h(e)\|_1 \leq 1$, Hogan, Scott and Tsarev measure how far G' is from representative of G by $f_h(G') = \|h(G') - \frac{e(G')}{e(G)}h(G)\|_1$, where $h(F) = \sum_{e \in E(F)} h(e)$. Their bounded-degree theorem gives $f_h(H) = O(k^2)$ for some Hamilton cycle H of K_N (and, by their methods, of $K_{n,n}$), and their Problem 6.4 asks for an improvement. We prove that $K_{n,n}$ ($n \geq 2$) and K_N ($N \geq 3$) always contain a Hamilton cycle H with $f_h(H) \leq 6k(\lceil \log_2 k \rceil + 2)$, respectively $f_h(H) \leq 6k(\lceil \log_2 k \rceil + 2) + 8k + 2$, i.e. $f_h(H) = O(k \log(k+1))$; the statement holds for labels in any k -dimensional normed space. The key step is a bound of $3k(\lceil \log_2 k \rceil + 2)$ for perfect matchings of $K_{n,n}$, which improves the paper's cornerstone Theorem 1.9 from $O(k^2)$ to $O(k \log k)$: two perfect matchings can be spliced into a third whose label sum is within $O(k)$ of their midpoint, and a dyadic averaging tree turns approximate midpoint closure into approximate convexity at the cost of only a logarithmic factor.

This note was produced by AI within the [Graph Theory AI](#) project: the argument was found by a language model and audited by an adversarial LLM referee, and it has not been checked by a human mathematician. [Section 5](#) records the full provenance.

1 Introduction

Representative subgraphs. Let G be a graph and $h: E(G) \rightarrow \mathbb{R}^k$ an assignment of vectors to its edges. For a subgraph $F \subseteq G$ write $h(F) = \sum_{e \in E(F)} h(e)$. Hogan, Scott and Tsarev [1] call $G' \subseteq G$ *representative* of G if $h(G')/e(G') = h(G)/e(G)$, and quantify the failure of this property by

$$f_h(G') = \left\| h(G') - \frac{e(G')}{e(G)} h(G) \right\|_1, \quad (1)$$

under the standing hypothesis $\|h(e)\|_1 \leq 1$ for all $e \in E(G)$ [1, §1]. (The paper's TeX source writes h through a macro named `\vec{h}`; the referee report in [Appendix B](#) accordingly writes $f_{\vec{h}}$ for the same quantity.) A k -edge-colouring c is the special case $h(e) = e_{c(e)}$ (a standard basis vector); then $h(G')$ is the vector of colour counts of G' , the paper writes f_c for f_h , and if G is colour-balanced (all colour classes of equal size) a representative subgraph is exactly a colour-balanced one.

Prior bounds. Pardey and Rautenbach [2] conjectured that every colour-balanced k -edge-coloured K_{2kt} has a perfect matching M with $f_c(M) = O(k^2)$, and proved $f_c(M) \leq 3k\sqrt{kt \log 2k}$; Hollom [3] removed the dependence on t with the bound $4k^2$; Hogan, Scott and Tsarev [1, Theorem 1.2] proved the conjecture, and more generally showed [1, Theorem 1.4] that for every n -vertex graph H of maximum degree Δ and every $h: E(K_n) \rightarrow \mathbb{R}^k$ with $\|h(e)\|_1 \leq 1$ there is a copy H' of H in K_n with $f_h(H') = O(\Delta^2 k^2 \sqrt{\log 2\Delta})$, answering a question of Banerjee and Hollom [4]. All of these results rest on one statement about complete bipartite graphs, which the authors call “a central tool, which we use throughout the paper” and “the cornerstone of our remaining results”:

Theorem 1.1 ([1, Theorem 1.9]). *Let $h: E(K_{n,n}) \rightarrow \mathbb{R}^k$ be such that $\|h(e)\|_1 \leq 1$ for all $e \in E(K_{n,n})$. Then there is a perfect matching M of $K_{n,n}$ satisfying $f_h(M) = O(k^2)$.*

It is proved in [1] via a linear relaxation and a necklace-splitting argument. In their concluding section the authors write: “Our use of [Theorem 1.9] to prove each of our other results for representative subgraphs implies that any improvement to the general k^2 bound in [Theorem 1.9] immediately gives an equal improvement to the k^2 factor in the bounds of each of our other results.” [1, §6] They then turn to Hamilton cycles: “since the maximum degree of a Hamilton cycle is 2, our general upper bound in [Theorem 1.4] implies that the error for embedding a representative Hamilton cycle in a complete graph is $O(k^2)$. Our methods imply that this bound also holds for Hamilton cycles in complete bipartite graphs.” They also observe that, by Alon’s necklace theorem [5], an almost representative Hamilton cycle with error ε yields an almost representative perfect matching with error $\varepsilon + O(k)$, “so any improvement to the k^2 upper bound for Hamilton cycles yields an improvement to the corresponding upper bound for perfect matchings”, and pose:

Problem 1.2 ([1, Problem 6.4]). Improve the bounds on $f_h(H)$ for a Hamilton cycle H in complete or complete bipartite graphs.

This note. We answer Problem 1.2 by lowering the bound from $O(k^2)$ to $O(k \log(k+1))$ in both hosts (Theorem 2.1 and Corollary 2.2). For a Hamilton cycle H one has $e(H)/e(K_{n,n}) = 2n/n^2 = 2/n$ and $e(H)/e(K_N) = N/\binom{N}{2} = 2/(N-1)$, so the centring vectors $\frac{2}{n}h(K_{n,n})$ and $\frac{2}{N-1}h(K_N)$ appearing in Theorem 2.1 are exactly the paper’s $\frac{e(H)}{e(G)}h(G)$: the theorem, specialised to $X = (\mathbb{R}^k, \|\cdot\|_1)$, is literally a bound on the quantity (1), with no reinterpretation. The argument runs in the direction opposite to the paper’s remark quoted above: we first improve the perfect-matching bound of Theorem 1.1 to $3k(\lceil \log_2 k \rceil + 2)$ (Proposition 2.4) and then reduce Hamilton cycles to perfect matchings of an auxiliary complete bipartite graph. By the authors’ own statement, the improvement of Theorem 1.1 propagates to the k^2 factor in each of their other results; we do not restate those consequences here.

The matching bound has two ingredients. A *splicing lemma* (Lemma 3.2): given two perfect matchings P, Q of a labelled $K_{r,r}$ there is a third, R , with $h(R)$ within $3Ld$ of $\frac{1}{2}(h(P) + h(Q))$, uniformly in r ; this uses the Hobby–Rice theorem [6, 7] to halve a vector-valued step function with at most d cuts. And a *dyadic convexification lemma* (Lemma 3.3): a finite set that is approximately closed under midpoints, with error D , lies within $D(\lceil \log_2 d \rceil + 2)$ of its convex hull in dimension d . Averaging along a balanced binary tree pays the error D at each level with geometrically decreasing weight, and only $O(\log d)$ levels contain nodes that are not constant; a sequential (Carathéodory-by-Carathéodory) rounding would instead pay $O(d)$ a total of $O(d)$ times and recover only $O(k^2)$. The referee could not locate Lemma 3.3 in the discrepancy literature, but could not exclude that it is folklore there (Remark 4.6).

The result was found in a second attempt on this problem; a first attempt by a different model, supplied as context, had reached the same strategy with weaker constants and without matching the paper’s own definition of f_h . Section 5 records what was inherited and what is new.

2 Main results

Throughout, X is a real normed space of finite dimension $d \geq 1$ with norm $\|\cdot\|$, $L > 0$, and $h: E(G) \rightarrow X$ satisfies $\|h(e)\| \leq L$ for all edges. We write $h(F) = \sum_{e \in E(F)} h(e)$ and put

$$B_d = \lceil \log_2 d \rceil + 2, \tag{2}$$

so that $B_1 = 2$ and $B_d = O(\log(d+1))$.

Theorem 2.1 (Main theorem). *Let X be a real normed space of dimension $d \geq 1$ and $h: E(G) \rightarrow X$ with $\|h(e)\| \leq L$ for every edge e .*

- (i) If $G = K_{n,n}$ with $n \geq 2$, then G has a Hamilton cycle H with $\|h(H) - \frac{2}{n}h(G)\| \leq 6LdB_d$.
- (ii) If $G = K_N$ with $N \geq 3$, then G has a Hamilton cycle H with $\|h(H) - \frac{2}{N-1}h(G)\| \leq 6LdB_d + 8Ld + 2L$.

In both cases the centring vector is the expected label sum of a uniformly random Hamilton cycle of G , and equals $\frac{e(H)}{e(G)}h(G)$.

Corollary 2.2 (Answer to [Problem 1.2](#)). Let $h: E(G) \rightarrow \mathbb{R}^k$ with $\|h(e)\|_1 \leq 1$ for all $e \in E(G)$, in the setting of [\[1\]](#). If $G = K_{n,n}$ with $n \geq 2$, some Hamilton cycle H satisfies $f_h(H) \leq 6k(\lceil \log_2 k \rceil + 2)$. If $G = K_N$ with $N \geq 3$, some Hamilton cycle H satisfies $f_h(H) \leq 6k(\lceil \log_2 k \rceil + 2) + 8k + 2$. In particular $f_h(H) = O(k \log(k+1))$ in both cases, improving the $O(k^2)$ bound of [\[1, Theorem 1.4 and §6\]](#).

Proof. Take $X = (\mathbb{R}^k, \|\cdot\|_1)$, $d = k$, $L = 1$ in [Theorem 2.1](#). Since $e(H) = 2n$, $e(K_{n,n}) = n^2$, $e(K_N) = \binom{N}{2}$ and a Hamilton cycle of K_N has N edges, $\frac{e(H)}{e(G)}h(G)$ equals $\frac{2}{n}h(K_{n,n})$, respectively $\frac{N}{\binom{N}{2}}h(K_N) = \frac{2}{N-1}h(K_N)$, so the left-hand sides in [Theorem 2.1](#) are exactly $f_h(H)$ as defined in [\(1\)](#). $\square$

Corollary 2.3 (Recolouring form). Let $k \geq 2$ and let c be a colour-balanced k -edge-colouring of the host graph. If the host is $K_{n,n}$ with $n \geq 2$ and $k \mid 2n$, some Hamilton cycle can be made colour-balanced by recolouring at most $3k(\lceil \log_2 k \rceil + 2)$ of its edges. If the host is K_N with $N \geq 3$ and $k \mid N$, some Hamilton cycle can be made colour-balanced by recolouring at most $3k(\lceil \log_2 k \rceil + 2) + 4k + 1$ of its edges.

Idea of proof. With $h(e) = e_{c(e)} \in (\mathbb{R}^k, \|\cdot\|_1)$ the centring vectors become $\frac{2n}{k}\mathbf{1}$ and $\frac{N}{k}\mathbf{1}$, and a subgraph F with kq edges and colour counts $m_1, \dots, m_k$ needs exactly $\sum_j (m_j - q)_+ = \frac{1}{2} \sum_j |m_j - q|$ recolourings to become colour-balanced. Halve the bounds of [Theorem 2.1](#). [\[full proof\]](#)

The substantive ingredient is the following bound for perfect matchings, which with $X = (\mathbb{R}^k, \|\cdot\|_1)$, $L = 1$ reads $f_h(M) \leq 3k(\lceil \log_2 k \rceil + 2)$ and thus improves [Theorem 1.1](#) from $O(k^2)$ to $O(k \log k)$.

Proposition 2.4 (Vector matching bound). Let $r \geq 1$ and $h: E(K_{r,r}) \rightarrow X$ with $\|h(e)\| \leq L$. Every point of the convex hull of $\{h(M) : M \text{ a perfect matching of } K_{r,r}\}$ lies within $3LdB_d$ of some $h(M)$. In particular there is a perfect matching M with

$$\|h(M) - \frac{1}{r}h(K_{r,r})\| \leq 3LdB_d.$$

Idea of proof. Let S be the set of perfect-matching label sums. [Lemma 3.2](#) says S is approximately midpoint-closed with error $D = 3Ld$; [Lemma 3.3](#) then puts every point of $\text{conv } S$ within DB_d of S . A uniformly random perfect matching contains each edge with probability $1/r$, so $\frac{1}{r}h(K_{r,r}) = \mathbb{E} h(M) \in \text{conv } S$. [\[full proof\]](#)

Idea of proof (of [Theorem 2.1](#)). (i) Order one side $A = \{a_1, \dots, a_n\}$ of $K_{n,n}$ cyclically and form an auxiliary $K_{n,n}$ between the n “gaps” $g_i = (a_i, a_{i+1})$ and the other side B , labelling $g_i b$ by $\lambda(g_i b) = h(a_i b) + h(a_{i+1} b)$, of norm at most $2L$. A perfect matching $g_i \mapsto b_i$ of the auxiliary graph is the Hamilton cycle $a_1 b_1 a_2 b_2 \dots a_n b_n a_1$, whose h -sum is the auxiliary λ -sum. Every edge $a_i b$ lies in exactly two labels, so the auxiliary centring $\frac{1}{n}\lambda(K_{n,n})$ is $\frac{2}{n}h(K_{n,n})$, and [Proposition 2.4](#) with $2L$ in place of L gives (i). (ii) For $N \geq 4$ first choose, by [Lemma 3.4](#), a near-bisection $V(K_N) = A \cup B$ with $|A| = \lceil N/2 \rceil = a$ whose crossing label sum $T = h(E(A, B))$ is within $2LdN$ of $p h(K_N)$, $p = 2ab/(N(N-1))$ being the crossing probability of a uniform edge. Run the gap construction between the cyclic gaps of A and B , adjoining a dummy vertex $*$ to B when N is odd with $\lambda(g_i *) = h(a_i a_{i+1})$ (the gap is then closed by the direct edge). Every auxiliary perfect matching is a Hamilton cycle of K_N . The auxiliary centring differs from $\frac{2}{N-1}h(K_N)$ by $\frac{2}{a}(T - p h(K_N))$ plus, for odd N , two vectors of norm at most L , since $2p/a$ equals $\frac{2}{N-1}$ (N even) or $\frac{2}{N}$ (N odd) and $\frac{2}{N} - \frac{2}{N-1} = -1/\binom{N}{2}$; this contributes at most $8Ld + 2L$. [Proposition 2.4](#) with $2L$ contributes $6LdB_d$. For $N = 3$ the unique Hamilton cycle is K_3 itself and the deviation is 0. [\[full proof\]](#)

3 The four lemmas

Lemma 3.1 (Hobby–Rice halving with d cuts). *Let $f: [0, r] \rightarrow \mathbb{R}^d$ be integrable. There is a set $A \subseteq [0, r]$, a union of intervals whose indicator is constant between at most d cut points, such that $\int_A f = \frac{1}{2} \int_0^r f$.*

Idea of proof. This is the theorem of Hobby and Rice [6]; the proof below is the Borsuk–Ulam proof of Alon and West [7]. For $x \in S^d \subset \mathbb{R}^{d+1}$ cut $[0, r]$ into consecutive intervals $J_0, \dots, J_d$ of lengths rx_i^2 and set $F(x) = \sum_i \text{sgn}(x_i) \int_{J_i} f$. Then $F: S^d \rightarrow \mathbb{R}^d$ is continuous and odd, so it vanishes somewhere; take A to be the union of the positively signed intervals. [\[full proof\]](#)

Lemma 3.2 (Splicing lemma). *Let $h: E(K_{r,r}) \rightarrow X$ with $\|h(e)\| \leq L$. For any two perfect matchings P, Q of $K_{r,r}$ there is a perfect matching R with*

$$\left\| h(R) - \frac{h(P) + h(Q)}{2} \right\| \leq 3Ld.$$

Idea of proof. Let the parts be U, V , view P, Q as bijections $V \rightarrow U$ and let σ be the permutation of V with $Q(v) = P(\sigma(v))$. List V as $v_1, \dots, v_r$ by concatenating the cycles of σ , and let P_i, Q_i be the P - and Q -edges at v_i . Apply [Lemma 3.1](#) to the step function equal to $h(P_i) - h(Q_i)$ on $[i-1, i]$, and let $\alpha_i \in [0, 1]$ be the measure of $A \cap [i-1, i]$; then $\sum_i \alpha_i (h(P_i) - h(Q_i)) = \frac{1}{2} \sum_i (h(P_i) - h(Q_i))$. Round α_i to the nearest $s_i \in \{0, 1\}$ and select P_i if $s_i = 1$, Q_i if $s_i = 0$. At most d of the α_i are fractional, so the selected edge set has label sum within $2L \cdot \frac{d}{2} = Ld$ of the midpoint. Each s_i is the value of $\mathbf{1}_A$ at a non-cut interior point of $[i-1, i]$, so the word $s_1 \dots s_r$ changes value at most d times, hence at most $2d$ times when each cycle block is read cyclically. In a block with $\sigma(v_i) = v_{i+1}$ the vertex $P(v_i) \in U$ has degree $1 + s_i - s_{i-1}$ in the selection, so there are exactly $b/2 \leq d$ vertices of U of degree 2 and as many of degree 0; delete one edge at each of the former and re-match the freed V -vertices to the latter, at cost at most $2Ld$. [\[full proof\]](#)

Lemma 3.3 (Dyadic convexification). *Let S be a finite subset of a d -dimensional normed space X such that for all $u, v \in S$ there is $w \in S$ with $\|w - \frac{u+v}{2}\| \leq D$. Then $\text{dist}(x, S) \leq DB_d$ for every $x \in \text{conv}(S)$. More precisely, if x is a convex combination of $m \geq 2$ points of S then $\text{dist}(x, S) \leq D(\lceil \log_2(m-1) \rceil + 2)$.*

Idea of proof. For dyadic coefficients with denominator 2^q , place the 2^q weighted copies of the m points in consecutive blocks at the leaves of a complete binary tree and assign to each node a representative in S : the common point if all leaves below are equal, otherwise an approximate midpoint of the children's representatives. The error at the root is at most $D \sum_j 2^{-j} H_j$, where $H_j \leq \min(2^j, m-1)$ counts the non-constant nodes at depth j (each contains one of the $m-1$ block boundaries), and $\sum_j \min(1, (m-1)2^{-j}) \leq \lceil \log_2(m-1) \rceil + 2$. Arbitrary coefficients are dyadic limits, and S is finite. Carathéodory gives $m \leq d+1$. [\[full proof\]](#)

Lemma 3.4 (Nearly representative bisection). *Let $N \geq 4$, $h: E(K_N) \rightarrow X$ with $\|h(e)\| \leq L$, $a = \lceil N/2 \rceil$, $b = \lfloor N/2 \rfloor$ and $p = \frac{2ab}{N(N-1)}$. There is a partition $V(K_N) = A \cup B$ with $|A| = a$, $|B| = b$ such that*

$$\|h(E(A, B)) - ph(K_N)\| \leq 2LdN.$$

Idea of proof. Choose A uniformly among a -sets and let I_e indicate that e crosses; $\mathbb{E}I_e = p$. For a scalar weight function $|w_e| \leq L$ the sum $Z = \sum_e w_e(I_e - p)$ has variance at most $\frac{17}{8}L^2N^2 < 4L^2N^2$: adjacent edges have $|\text{Cov}(I_e, I_f)| = p|\frac{1}{2} - p| \leq \frac{1}{N}$, disjoint edges have $\text{Cov}(I_e, I_f) \leq \frac{4}{N^2}$, and there are $\binom{N}{2}$ edges, $3\binom{N}{3}$ adjacent and $3\binom{N}{4}$ disjoint pairs. Pass from scalars to X through an Auerbach basis $u_1, \dots, u_d$ with coordinate functionals φ_j of dual norm one [8, Prop. 1.c.3], so that $\|x\| \leq \sum_j |\varphi_j(x)|$; then $\mathbb{E}\|\sum_e (I_e - p)h(e)\| \leq \sum_j \sqrt{\text{Var } \varphi_j(\cdot)} \leq 2LdN$. [\[full proof\]](#)

4 Remarks

Remark 4.1 (Interpretation). The referee downloaded the TeX source of arXiv:2604.09449v2 (posted 11 July 2026, the current version) and checked that (1) and Problem 1.2 are quoted exactly, that the paper’s baseline for Hamilton cycles is $O(k^2)$ (Theorem 1.4 with $\Delta = 2$, extended to $K_{n,n}$ “by our methods”), and that the centring vectors of Theorem 2.1 coincide with $\frac{e(H)}{e(G)}h(G)$ as computed in the proof of Corollary 2.2. The hypothesis $\|h(e)\| \leq L$ specialises to the paper’s $\|h(e)\|_1 \leq 1$. No sub-class of instances is excluded: the divisibility conditions $k \mid 2n, k \mid N$ appear only in Corollary 2.3, where they are necessary for a colour-balanced Hamilton cycle to exist at all. Problem 1.2 asks for an improvement, not for the optimal order, and the note does not determine the order (Remark 4.3).

Remark 4.2 (The matching bound and its consequences). Proposition 2.4 with $X = (\mathbb{R}^k, \|\cdot\|_1)$ and $L = 1$ gives a perfect matching M of $K_{n,n}$ with $f_h(M) \leq 3k(\lceil \log_2 k \rceil + 2)$, improving Theorem 1.1 from $O(k^2)$ to $O(k \log k)$. The authors of [1] state that any improvement of the k^2 bound in that theorem “immediately gives an equal improvement to the k^2 factor in the bounds of each of our other results” (§6; quoted in the Introduction). Taken at their word, the k^2 factors in the paper’s Theorems 1.2–1.4 and its further results become $k \log k$. We have not re-derived those propagated statements and do not claim them here; a reader acting on this note should check each reduction in [1] for uses of the theorem beyond its statement.

Remark 4.3 (Order of magnitude and lower bounds). Nothing here rules out an $O(k)$ bound for Hamilton cycles. For colourings, [1] constructs colour-balanced k -edge-colourings of $K_{kt,kt}$ in which every perfect matching has $f_c(M) \geq \sqrt{k/2}$ and conjectures (Conjecture 6.1) that $O(\sqrt{k})$ is attainable for colour-balanced perfect matchings; that conjecture is untouched by this note. In the general vector setting the situation is different: the sibling target 2604.09449__00 of this project exhibits labellings $h: E(K_{n,n}) \rightarrow \mathbb{R}^k$ with $\|h(e)\|_1 \leq 1$ for which every perfect matching has $f_h(M) = k$. The referee re-ran that construction and found $f_h(M) = k$ exactly for $k = 2, 4, 6, 8$, against the bounds $3kB_k = 18, 48, 90, 120$ of Proposition 2.4; so in the vector setting the matching bound is tight up to the factor B_k . The constants have not been optimised.

Remark 4.4 (Which Hamilton cycles). The proof of Theorem 2.1(ii) only ever produces Hamilton cycles that alternate between the two sides of a near-bisection $A \cup B$ (with one edge inside A when N is odd), and (i) only produces cycles in which the vertices of one side appear in a prescribed cyclic order. This is legitimate for an upper bound, but the argument says nothing about Hamilton cycles outside these families.

Remark 4.5 (Computational checks by the referee). The scripts are in `verification_astra/scripts/2604.09449__03/`. (a) *Lemma 3.4, exact.* Enumerating all $\binom{N}{a}$ partitions with rational arithmetic for $N = 3, \dots, 10$: $\mathbb{P}(I_e = 1) = p$, $\mathbb{P}(I_e = I_f = 1) = p/2$ for adjacent e, f , and the two closed forms for $\text{Cov}(I_e, I_f)$ on disjoint pairs hold as exact identities, with $(p, \text{Cov}_{\text{disj}}) = (\frac{2}{3}, \frac{2}{9}), (\frac{3}{5}, \frac{1}{25}), (\frac{3}{5}, \frac{1}{25}), (\frac{4}{7}, \frac{4}{245}), (\frac{4}{7}, \frac{4}{245}), (\frac{5}{9}, \frac{5}{567}), (\frac{5}{9}, \frac{5}{567})$ for $N = 4, \dots, 10$; the inequality $\text{Cov}_{\text{disj}} \leq 4/N^2$ is tightest at $N = 4$ ($0.222 \leq 0.25$). The exact value of $\sum_{e,f} |\text{Cov}(I_e, I_f)|$ is 5.33, 7.20, 14.40, 17.14, 27.43, 31.11 for $N = 4, \dots, 9$, against the bounding expression 9.00, 19.29, 33.75, 52.39, 75.25, 102.33 used in the proof. (b) *Exhaustive checks* of the ratio (best attainable value)/(claimed bound), which must be at most 1: Lemma 3.2 over random ℓ_1 -normalised labellings, 60 instances each, ratios 0.048–0.140 for $(r, d) \in \{(5, 2), (5, 3), (6, 2), (6, 3), (6, 4), (7, 3)\}$; Proposition 2.4 ratios 0.007–0.019 on the same instances; Theorem 2.1(i) over all Hamilton cycles of $K_{n,n}$: 0.034, 0.015, 0.012, 0.006 for $(n, d) = (4, 2), (5, 2), (5, 3), (6, 3)$; Theorem 2.1(ii) over all Hamilton cycles of K_N : 0.019, 0.014, 0.007, 0.004 for $(N, d) = (5, 2), (6, 2), (6, 3), (7, 3)$. (c) *Structure.* For $N = 3, \dots, 10$ every one of the $a!$ auxiliary perfect matchings in the proof of (ii), dummy vertex included for odd N , yields a genuine Hamilton cycle of K_N ; likewise for $K_{n,n}$. (d) *Adversarial search.* Hill-climbing over labellings to maximise the same ratios never exceeded 0.333 (Lemma 3.2 at $(r, d) = (4, 1)$); Lemma 3.1 was confirmed numerically for $d = 2, r = 6$ on random step functions. No computational claim failed.

Remark 4.6 (Novelty). The referee found no prior occurrence of the result: Semantic Scholar lists no citations of arXiv:2604.09449; an arXiv search for “colour-balanced”/“color-balanced” returns, for 2026, only the source paper and an unrelated paper on colour-biased tight Hamilton cycles; the known chain of matching bounds is $3k\sqrt{kt \log 2k}$ [2] $\rightarrow 4^{k^2}$ [3] $\rightarrow O(k^2)$ [1]; and v2 of the source paper still poses [Problem 1.2](#) with the $O(k^2)$ baseline. The only web page connecting an $O(k \log k)$ bound to the paper is this project’s own repository, which is self-contamination and not prior art. What the referee could *not* certify is the novelty of [Lemma 3.3](#): the statement that an approximately midpoint-closed finite set lies within $D(\lceil \log_2 d \rceil + 2)$ of its convex hull is general enough that it may exist, in other language, in the discrepancy literature on rounding and vector balancing (Beck–Fiala [9], Steinitz-type results of Bárány and Grinberg [10], Banaszczyk [11]). The referee checked that the two obvious routes do not yield [Proposition 2.4](#) directly, since the constraints involve every edge and $\{0, 1\}$ -rounding of a fractional matching does not respect the matching structure, and records this as a novelty caveat, not a correctness one. A human expert should search that literature before any submission.

Remark 4.7 (The sibling target). The catalog entry 2604.09449__00 (same paper, first campaign) was rated `ALREADY_KNOWN` by the referee of that campaign: its counterexample refuted the vector-label form of Conjecture 6.1 as stated in arXiv:2604.09449v1, which the authors had already replaced in v2 (11 July 2026) by the colour-balanced form quoted in [Remark 4.3](#), acknowledging V. Rozhoň for pointing out the error. That target is relevant here only as the source of the $\Omega(k)$ labelling in [Remark 4.3](#); the present problem is Problem 6.4, which is open in v2.

5 Provenance

The mathematics in this note was produced without human intervention, in three stages.

(1) The argument was found by the language model `gpt-6-astra` (reasoning effort *max*, single autonomous pass, started 2026-09-17) as part of the *retry* leg of a sweep over open problems catalogued from arXiv (catalog id 2604.09449__03, leg `attacks_retry`; writeup in `attacks_retry/2604.09449__03/output.md`). This was a *second* attempt on the problem. The first, by `gpt-5.6-sol` (`attacks/2604.09449__03`, self-assessed *partial*), was supplied to the second model as context, without a referee report. The overall strategy is inherited from that first attempt: the reduction of Hamilton cycles to perfect matchings of an auxiliary “gap” graph, the necklace-halving splicing of two matchings, the dyadic averaging tree, and the random near-bisection of K_N all appear there, with the bounds $8Ld$ for the splicing lemma and $8k(\lceil \log_2 k \rceil + 2)(+4k + 1)$ for the recolouring form, and only in the ℓ_1 colouring setting with f_h guessed rather than taken from the paper. The second writeup re-verified those steps and improved them in three ways. It realises each rounded bit s_i in the splicing lemma as the value of $\mathbf{1}_A$ at a non-cut interior point, which makes the bound of d on the number of changes of $s_1 \cdots s_r$ immediate and sharpens the splicing constant from $8Ld$ to $3Ld$ (the first attempt chose s_i arbitrarily on the intervals containing a cut and then argued, less transparently, that each such choice creates at most two further changes, reaching $6d$ cyclic changes instead of $2d$). It generalises the bisection lemma from ℓ_1 colour counts to an arbitrary norm through an Auerbach basis. And it states the theorem in the paper’s own vector-label formulation with the paper’s centring vectors, so that the result is a bound on f_h rather than on a recolouring statistic.

(2) An independent adversarial referee agent (`claude-opus-5`, 2026-09-17) audited the writeup against the TeX source of arXiv:2604.09449v2. It confirmed the interpretation ([Remark 4.1](#)), re-derived every step it isolated (twenty-one, all rated valid, none a gap), identified [Lemma 3.1](#) as the Hobby–Rice theorem, checked [Lemma 3.4](#) exactly for $N \leq 10$ and [Theorem 2.1](#) exhaustively for $K_{n,n}$ up to $n = 6$ and K_N up to $N = 7$ ([Remark 4.5](#)), searched for prior art ([Remark 4.6](#)), and noted that the writeup understates its result by treating [Proposition 2.4](#) as an aside ([Remark 4.2](#)). Its verdict was `CONFIRMED`. Its report is reproduced verbatim in [Appendix B](#).

(3) On 2026-09-17 the writeup was rewritten into the present note by `claude-fable-5-1`. The text was reorganised, with proof ideas in the main text and full proofs in [Appendix A](#). Every deviation from

the writeup is listed here. *No mathematical statement or proof was changed.* (a) Notation: the paper’s f_h is used throughout (the writeup wrote f_h in its conclusion and the referee wrote $f_{\vec{h}}$; both denote (1)). (b) [Corollary 2.2](#) was added: it is the writeup’s Theorem 1 specialised to $X = (\mathbb{R}^k, \|\cdot\|_1)$, $d = k$, $L = 1$, an instantiation the writeup states in words (“in particular, for unit-bounded ...labels”) but not with explicit constants. (c) Citations were added: Hobby–Rice [6] and Alon–West [7] for [Lemma 3.1](#), which the writeup proved without attribution; Lindenstrauss–Tzafriri [8] for the Auerbach basis; Alon [5] where the source paper invokes it; Pardey–Rautenbach [2], Hollom [3] and Banerjee–Hollom [4] for the history; Beck–Fiala [9], Bárány–Grinberg [10] and Banaszczyk [11] for the novelty caveat. (d) Theorem and problem numbers of [1] (1.2, 1.4, 1.9, Conjecture 6.1, Problem 6.4) were read off the v2 TeX source; the catalog page cited “Theorem 1.2” for the bounded-degree bound, which is its v1 number. (e) Routine steps that the writeup left terse were spelled out in [Appendix A](#), following the referee’s re-derivations: the continuity of the map in [Lemma 3.1](#); the one-line computations of $\mathbb{P}(I_e = I_f = 1) = p/2$, of the two closed-form covariances and of the inequality $\text{Cov} \leq 4/N^2$ in [Lemma 3.4](#); the recursion behind the error bound and the explicit dyadic approximation in [Lemma 3.3](#); the construction of the Auerbach basis; and the verification that the auxiliary matchings are Hamilton cycles. (f) [Section 4](#) is drawn from the referee report and from the source paper’s Section 6.

No human mathematician has checked this argument. We circulate it for human review, not as a claim to a new result.

A Full proofs

Throughout, X , d , L , h and B_d are as in [Section 2](#).

Lemma 3.1 (Hobby–Rice halving with d cuts). *Let $f: [0, r] \rightarrow \mathbb{R}^d$ be integrable. There is a set $A \subseteq [0, r]$, a union of intervals whose indicator is constant between at most d cut points, such that $\int_A f = \frac{1}{2} \int_0^r f$.*

Proof. Let $S^d = \{x \in \mathbb{R}^{d+1} : \sum_i x_i^2 = 1\}$. For $x = (x_0, \dots, x_d) \in S^d$ let $J_0(x), \dots, J_d(x)$ be the consecutive closed intervals partitioning $[0, r]$ with $|J_i(x)| = rx_i^2$, i.e. $J_i(x) = [t_i(x), t_{i+1}(x)]$ with $t_i(x) = r \sum_{l < i} x_l^2$, and define

$$F(x) = \sum_{i=0}^d \text{sgn}(x_i) \int_{J_i(x)} f(t) dt \in \mathbb{R}^d.$$

F is odd: $J_i(-x) = J_i(x)$ while every sign flips. F is continuous: the endpoints $t_i(x)$ depend continuously on x , and $x \mapsto \int_{t_i(x)}^{t_{i+1}(x)} f$ is continuous because f is integrable (absolute continuity of the integral). The only discontinuities of $\text{sgn}(x_i)$ occur where $x_i = 0$; there $|J_i(x)| = rx_i^2 \rightarrow 0$, so the i -th summand tends to 0 regardless of the sign, and F is continuous there as well. By the Borsuk–Ulam theorem an odd continuous map $S^d \rightarrow \mathbb{R}^d$ has a zero x^* . Let A be the union of the $J_i(x^*)$ with $x_i^* > 0$. Intervals with $x_i^* = 0$ are degenerate, so $F(x^*) = 0$ reads $\int_A f - \int_{[0,r] \setminus A} f = 0$, i.e. $\int_A f = \frac{1}{2} \int_0^r f$. Since A is a union of some of the $d+1$ consecutive intervals, $\mathbf{1}_A$ is constant between the at most d internal endpoints $t_1(x^*), \dots, t_d(x^*)$. $\square$

Lemma 3.2 (Splicing lemma). *Let $h: E(K_{r,r}) \rightarrow X$ with $\|h(e)\| \leq L$. For any two perfect matchings P, Q of $K_{r,r}$ there is a perfect matching R with*

$$\left\| h(R) - \frac{h(P) + h(Q)}{2} \right\| \leq 3Ld.$$

Proof. Let the parts of $K_{r,r}$ be U and V . Regard P and Q as bijections $V \rightarrow U$ and define the permutation σ of V by $Q(v) = P(\sigma(v))$. Write $V = \{v_1, \dots, v_r\}$ by listing the cycles of σ one after another, each in

its cyclic order; so within each block (cycle) $\sigma(v_i) = v_{i+1}$, indices taken cyclically inside the block. Let $P_i = v_i P(v_i)$ and $Q_i = v_i Q(v_i) = v_i P(v_{i+1})$ be the P - and Q -edges at v_i .

Halving. Identify X linearly with $\mathbb{R}^d$ and let $f: [0, r] \rightarrow \mathbb{R}^d$ be the step function equal to $h(P_i) - h(Q_i)$ on $I_i = [i-1, i]$. Lemma 3.1 gives $A \subseteq [0, r]$ with at most d cut points and $\int_A f = \frac{1}{2} \int_0^r f$. With $\alpha_i = |A \cap I_i| \in [0, 1]$ this reads

$$\sum_{i=1}^r \alpha_i (h(P_i) - h(Q_i)) = \frac{1}{2} \sum_{i=1}^r (h(P_i) - h(Q_i)) = \frac{h(P) - h(Q)}{2}. \quad (3)$$

Let $s_i \in \{0, 1\}$ be the value nearest to α_i (ties broken arbitrarily). If $0 < \alpha_i < 1$ then $\mathbf{1}_A$ is not constant on I_i , so the open interval $(i-1, i)$ contains a cut point; these open intervals are disjoint, so at most d indices have $\alpha_i \notin \{0, 1\}$, and each contributes at most $\frac{1}{2}$ to

$$\sum_i |s_i - \alpha_i| \leq \frac{d}{2}. \quad (4)$$

Select the edge P_i if $s_i = 1$ and Q_i if $s_i = 0$, and let z be the label sum of the selected edges. Then $z = \sum_i (s_i h(P_i) + (1 - s_i) h(Q_i))$, so by (3)

$$z - \frac{h(P) + h(Q)}{2} = \sum_i \left(s_i - \frac{1}{2}\right) (h(P_i) - h(Q_i)) = \sum_i (s_i - \alpha_i) (h(P_i) - h(Q_i)),$$

and by (4) and $\|h(P_i) - h(Q_i)\| \leq 2L$,

$$\left\| z - \frac{h(P) + h(Q)}{2} \right\| \leq 2L \cdot \frac{d}{2} = Ld. \quad (5)$$

Counting changes. For each i choose a point t_i in the interior of I_i that is not a cut point and satisfies $\mathbf{1}_A(t_i) = s_i$: if $\alpha_i \in \{0, 1\}$ then $\mathbf{1}_A = s_i$ on I_i except possibly at cut points; if $0 < \alpha_i < 1$ both values are taken on sets of positive measure in I_i . Then $t_1 < t_2 < \dots < t_r$, and whenever $s_i \neq s_{i+1}$ the interval (t_i, t_{i+1}) contains a cut point. These intervals are disjoint, so the linear word $s_1 s_2 \dots s_r$ has at most d changes. Now read each block cyclically. If a block has c internal (linear) changes, closing it adds one change exactly when its first and last letters differ, which forces $c \geq 1$; so the block has at most $2c$ cyclic changes. The internal changes of the blocks are among the at most d linear changes (a change across a block boundary is internal to no block), hence the total number b of cyclic changes over all blocks satisfies

$$b \leq 2d. \quad (6)$$

Degrees. Every $v_i \in V$ has exactly one selected edge. A vertex $u \in U$ equals $P(v_i)$ for exactly one i ; inside the block of v_i the selected edges at u are P_i (present iff $s_i = 1$) and $Q_{i-1} = v_{i-1} P(v_i)$ (present iff $s_{i-1} = 0$), indices cyclic in the block. Hence

$$\deg(P(v_i)) = s_i + (1 - s_{i-1}) = 1 + s_i - s_{i-1} \in \{0, 1, 2\},$$

and $P(v_i)$ has degree 2 exactly at the cyclic changes $s_{i-1} s_i = 01$ and degree 0 exactly at the changes 10. In a cyclic binary word the two kinds of changes are equinumerous, so there are exactly $b/2$ vertices of U of degree 2 and $b/2$ of degree 0.

Repair. At each degree-2 vertex of U delete one of its two selected edges. The deleted edges have distinct endpoints in V (each V -vertex carried one selected edge), so exactly $b/2$ vertices of V lose their edge; match them arbitrarily to the $b/2$ degree-0 vertices of U , which is possible because the host is complete bipartite. The result is a perfect matching R obtained from the selection by exchanging $b/2 \leq d$ edges, so

$$\|h(R) - z\| \leq 2L \cdot \frac{b}{2} \leq 2Ld. \quad (7)$$

Combining (5) and (7) gives the lemma. The bound does not depend on r . $\square$

Lemma 3.3 (Dyadic convexification). *Let S be a finite subset of a d -dimensional normed space X such that for all $u, v \in S$ there is $w \in S$ with $\|w - \frac{u+v}{2}\| \leq D$. Then $\text{dist}(x, S) \leq DB_d$ for every $x \in \text{conv}(S)$. More precisely, if x is a convex combination of $m \geq 2$ points of S then $\text{dist}(x, S) \leq D(\lceil \log_2(m-1) \rceil + 2)$.*

Proof. Let $x = \sum_{i=1}^m \beta_i x_i$ with distinct $x_i \in S$, $\beta_i > 0$, $\sum_i \beta_i = 1$. If $m = 1$ then $x \in S$. Assume $m \geq 2$.

Dyadic coefficients. Suppose first that every β_i is a multiple of 2^{-q} . Take a complete binary tree of depth q with 2^q leaves, and label the leaves, from left to right, with $\beta_1 2^q$ copies of x_1 , then $\beta_2 2^q$ copies of x_2 , and so on (the m blocks). For a node v at depth j let $\Lambda(v)$ be its 2^{q-j} leaves and $x_v = 2^{-(q-j)} \sum_{\ell \in \Lambda(v)} y_\ell$ the average of their labels; then $x_{\text{root}} = x$ and $x_v = \frac{1}{2}(x_{v_1} + x_{v_2})$ for the children v_1, v_2 of v . Call v *constant* if all labels in $\Lambda(v)$ coincide. Assign representatives $z_v \in S$ bottom-up: for a leaf, $z_v = y_v$; for a constant internal node, z_v is the common label (so $z_v = x_v$); otherwise choose $z_v \in S$ with $\|z_v - \frac{1}{2}(z_{v_1} + z_{v_2})\| \leq D$, using the hypothesis. For a non-constant node,

$$\|z_v - x_v\| \leq \left\| z_v - \frac{z_{v_1} + z_{v_2}}{2} \right\| + \frac{1}{2}(\|z_{v_1} - x_{v_1}\| + \|z_{v_2} - x_{v_2}\|) \leq D + \frac{1}{2}(\|z_{v_1} - x_{v_1}\| + \|z_{v_2} - x_{v_2}\|),$$

while for a constant node $\|z_v - x_v\| = 0$. Unrolling from the root, and letting H_j be the number of non-constant nodes at depth j ,

$$\|z_{\text{root}} - x\| \leq D \sum_{j=0}^{q-1} 2^{-j} H_j.$$

Trivially $H_j \leq 2^j$. Moreover, a non-constant node's leaves form an interval of leaves containing two different labels, hence containing one of the $m-1$ boundaries between consecutive blocks; leaf intervals of distinct nodes at the same depth are disjoint, so $H_j \leq m-1$. With $t = \lceil \log_2(m-1) \rceil$, so that $2^t \geq m-1$,

$$\sum_{j \geq 0} \min(1, (m-1)2^{-j}) \leq \sum_{j < t} 1 + \sum_{j \geq t} (m-1)2^{-j} \leq t + (m-1)2^{-t} \cdot 2 \leq t + 2.$$

Hence $\text{dist}(x, S) \leq \|z_{\text{root}} - x\| \leq D(\lceil \log_2(m-1) \rceil + 2)$.

Arbitrary coefficients. For $q \geq 1$ let $\beta_i^{(q)}$ be dyadic rationals with denominator 2^q , $\sum_i \beta_i^{(q)} = 1$, $\beta_i^{(q)} \geq 0$ and $\beta_i^{(q)} \rightarrow \beta_i$ as $q \rightarrow \infty$ (round each β_i down to a multiple of 2^{-q} and add the total deficit, which is less than $m2^{-q}$, to $\beta_1^{(q)}$). Put $x^{(q)} = \sum_i \beta_i^{(q)} x_i \rightarrow x$. Dropping the indices with $\beta_i^{(q)} = 0$ leaves $m' \leq m$ points, and the dyadic case yields $z_q \in S$ with $\|z_q - x^{(q)}\| \leq D(\lceil \log_2(m-1) \rceil + 2)$ (for $m' = 1$, $z_q = x^{(q)}$). As S is finite, some $z \in S$ equals z_q for infinitely many q ; letting $q \rightarrow \infty$ along these, $\|z - x\| \leq D(\lceil \log_2(m-1) \rceil + 2)$.

Conclusion. By Carathéodory's theorem every $x \in \text{conv}(S)$ is a convex combination of at most $m \leq d+1$ points of S , and $\lceil \log_2(m-1) \rceil + 2 \leq \lceil \log_2 d \rceil + 2 = B_d$. $\square$

Proposition 2.4 (Vector matching bound). *Let $r \geq 1$ and $h: E(K_{r,r}) \rightarrow X$ with $\|h(e)\| \leq L$. Every point of the convex hull of $\{h(M) : M \text{ a perfect matching of } K_{r,r}\}$ lies within $3LdB_d$ of some $h(M)$. In particular there is a perfect matching M with*

$$\|h(M) - \frac{1}{r} h(K_{r,r})\| \leq 3LdB_d.$$

Proof. Let $S = \{h(M) : M \text{ a perfect matching of } K_{r,r}\}$, a finite subset of X . By Lemma 3.2, for all $u = h(P), v = h(Q)$ in S there is $w = h(R) \in S$ with $\|w - \frac{u+v}{2}\| \leq 3Ld$. Lemma 3.3 with $D = 3Ld$ gives $\text{dist}(x, S) \leq 3LdB_d$ for every $x \in \text{conv}(S)$, which is the first assertion. For the second, let M be a uniformly random perfect matching of $K_{r,r}$; each edge lies in $(r-1)!$ of the $r!$ perfect matchings, so $\mathbb{P}(e \in M) = 1/r$ and

$$\frac{1}{r} h(K_{r,r}) = \sum_e \mathbb{P}(e \in M) h(e) = \mathbb{E} h(M) = \sum_M \frac{1}{r!} h(M) \in \text{conv}(S). \quad \square$$

Lemma 3.4 (Nearly representative bisection). *Let $N \geq 4$, $h: E(K_N) \rightarrow X$ with $\|h(e)\| \leq L$, $a = \lceil N/2 \rceil$, $b = \lfloor N/2 \rfloor$ and $p = \frac{2ab}{N(N-1)}$. There is a partition $V(K_N) = A \cup B$ with $|A| = a$, $|B| = b$ such that*

$$\|h(E(A, B)) - p h(K_N)\| \leq 2LdN.$$

Proof. Let A be a uniformly random a -subset of $V = V(K_N)$, $B = V \setminus A$, and for an edge e let $I_e = \mathbf{1}[e \text{ has one end in } A \text{ and one in } B]$. There are ab crossing edges out of $\binom{N}{2}$, so by symmetry $\mathbb{E}I_e = ab/\binom{N}{2} = p$.

Scalar variance bound. Let w_e be real weights with $|w_e| \leq L$ and $Z = \sum_e w_e(I_e - p)$. Then $\text{Var } Z = \sum_{e,f} w_e w_f \text{Cov}(I_e, I_f) \leq L^2(\sum_e \text{Var } I_e + \sum_{e \neq f} |\text{Cov}(I_e, I_f)|)$, and $\text{Var } I_e = p(1-p) \leq \frac{1}{4}$.

For adjacent edges $e = xy$, $f = xz$: both cross iff $x \in A$, $y, z \in B$ or $x \in B$, $y, z \in A$. Revealing the sides of x, y, z in turn,

$$\mathbb{P}(I_e = I_f = 1) = \frac{ab(b-1) + ba(a-1)}{N(N-1)(N-2)} = \frac{ab(a+b-2)}{N(N-1)(N-2)} = \frac{ab}{N(N-1)} = \frac{p}{2},$$

so $\text{Cov}(I_e, I_f) = \frac{p}{2} - p^2 = p(\frac{1}{2} - p)$. If N is even, $p = \frac{N}{2(N-1)}$ and $\frac{1}{2} - p = -\frac{1}{2(N-1)}$; if N is odd, $p = \frac{N+1}{2N}$ and $\frac{1}{2} - p = -\frac{1}{2N}$. As $p \leq 1$, in both cases $|\text{Cov}(I_e, I_f)| \leq \frac{1}{2(N-1)} \leq \frac{1}{N}$.

For disjoint edges $e = xy$, $f = zw$: both cross iff exactly one of x, y and exactly one of z, w lies in A , which happens in 4 patterns, each of probability $\frac{a(a-1)b(b-1)}{N(N-1)(N-2)(N-3)}$. Hence $\mathbb{P}(I_e = I_f = 1) = \frac{4a(a-1)b(b-1)}{N(N-1)(N-2)(N-3)}$, and subtracting p^2 ,

$$\text{Cov}(I_e, I_f) = \begin{cases} \frac{N}{2(N-1)^2(N-3)} & N \text{ even,} \\ \frac{N+1}{2N^2(N-2)} & N \text{ odd.} \end{cases}$$

(For N even, $a = b = \frac{N}{2}$ gives $\mathbb{P} = \frac{N(N-2)}{4(N-1)(N-3)}$ and $p^2 = \frac{N^2}{4(N-1)^2}$; for N odd, $a = \frac{N+1}{2}$, $b = \frac{N-1}{2}$ give $\mathbb{P} = \frac{(N+1)(N-1)}{4N(N-2)}$ and $p^2 = \frac{(N+1)^2}{4N^2}$; the differences simplify as displayed.) Both are at most $\frac{4}{N^2}$ for $N \geq 4$: for even N this is $N^3 \leq 8(N-1)^2(N-3)$, true at $N = 4$ ($64 \leq 72$) and for larger even N since the right side grows faster; for odd N it is $N+1 \leq 8(N-2)$, true for $N \geq 3$.

There are $\binom{N}{2}$ edges, $3\binom{N}{3}$ unordered adjacent pairs and $3\binom{N}{4}$ unordered disjoint pairs, so

$$\begin{aligned} \text{Var } Z &\leq L^2 \left(\frac{1}{4} \binom{N}{2} + \frac{6}{N} \binom{N}{3} + \frac{24}{N^2} \binom{N}{4} \right) \\ &\leq L^2 \left(\frac{N^2}{8} + (N-1)(N-2) + \frac{(N-1)(N-2)(N-3)}{N} \right) \\ &\leq \frac{17}{8} L^2 N^2 < 4L^2 N^2. \end{aligned}$$

From scalars to X . Fix any linear identification $X \cong \mathbb{R}^d$ and choose $u_1, \dots, u_d$ in the unit ball of X maximising $|\det(u_1, \dots, u_d)|$; the maximum exists by compactness and is positive. Let φ_j be the coordinate functionals, $\varphi_j(x) = \det(u_1, \dots, u_{j-1}, x, u_{j+1}, \dots, u_d) / \det(u_1, \dots, u_d)$ by Cramer's rule. For $\|x\| \leq 1$ maximality gives $|\varphi_j(x)| \leq 1$, so φ_j has dual norm at most 1 (in fact 1, as $\varphi_j(u_j) = 1$, which also forces $\|u_j\| = 1$). This is an Auerbach basis [8, Prop. 1.c.3], and $x = \sum_j \varphi_j(x) u_j$ gives

$$\|x\| \leq \sum_{j=1}^d |\varphi_j(x)| \quad (x \in X).$$

Let $T = \sum_e (I_e - p)h(e) = h(E(A, B)) - p h(K_N)$. For each j , $\varphi_j(T) = \sum_e (I_e - p)\varphi_j(h(e))$ is a scalar sum Z with weights $|\varphi_j(h(e))| \leq \|h(e)\| \leq L$ and mean 0, so by Cauchy-Schwarz

$\mathbb{E}|\varphi_j(T)| \leq \sqrt{\text{Var } \varphi_j(T)} \leq 2LN$. Therefore $\mathbb{E}\|T\| \leq \sum_j \mathbb{E}|\varphi_j(T)| \leq 2LdN$, and some partition (A, B) satisfies $\|T\| \leq 2LdN$. $\square$

Theorem 2.1 (Main theorem). *Let X be a real normed space of dimension $d \geq 1$ and $h: E(G) \rightarrow X$ with $\|h(e)\| \leq L$ for every edge e .*

- (i) *If $G = K_{n,n}$ with $n \geq 2$, then G has a Hamilton cycle H with $\|h(H) - \frac{2}{n}h(G)\| \leq 6LdB_d$.*
- (ii) *If $G = K_N$ with $N \geq 3$, then G has a Hamilton cycle H with $\|h(H) - \frac{2}{N-1}h(G)\| \leq 6LdB_d + 8Ld + 2L$.*

In both cases the centring vector is the expected label sum of a uniformly random Hamilton cycle of G , and equals $\frac{e(H)}{e(G)}h(G)$.

Proof. (i) $G = K_{n,n}$, $n \geq 2$. Let the parts be $A = \{a_1, \dots, a_n\}$, cyclically ordered with $a_{n+1} = a_1$, and B . Form the auxiliary complete bipartite graph K^{aux} between the n gaps $g_i = (a_i, a_{i+1})$ and B , with labels

$$\lambda(g_i b) = h(a_i b) + h(a_{i+1} b), \quad \|\lambda(g_i b)\| \leq 2L.$$

A perfect matching $g_i \mapsto b_i$ of K^{aux} assigns distinct $b_i \in B$ to the gaps, and $H = a_1 b_1 a_2 b_2 \dots a_n b_n a_1$ is a Hamilton cycle of $K_{n,n}$ (it visits all $2n$ vertices once; for $n = 2$ it is the unique 4-cycle) with $h(H) = \sum_i (h(a_i b_i) + h(a_{i+1} b_i)) = \lambda(\text{matching})$. Each edge $a_i b$ occurs in exactly two labels, $\lambda(g_{i-1} b)$ and $\lambda(g_i b)$, so $\lambda(K^{\text{aux}}) = 2h(K_{n,n})$ and $\frac{1}{n}\lambda(K^{\text{aux}}) = \frac{2}{n}h(K_{n,n})$. Proposition 2.4 applied to K^{aux} with label bound $2L$ gives a perfect matching, hence a Hamilton cycle H , with $\|h(H) - \frac{2}{n}h(K_{n,n})\| \leq 3(2L)dB_d = 6LdB_d$.

(ii) $G = K_N$. For $N = 3$ the only Hamilton cycle is K_3 itself and $\frac{2}{N-1}h(K_3) = h(K_3)$, so the deviation is 0. Let $N \geq 4$, and let $A \cup B$ be the partition of Lemma 3.4, $|A| = a = \lceil N/2 \rceil$, $|B| = b$. Write $S = h(K_N)$ and $T = h(E(A, B))$, so $\|T - pS\| \leq 2LdN$. Order $A = \{a_1, \dots, a_a\}$ cyclically. If N is even, $b = a$; if N is odd, $b = a - 1$ and we adjoin a dummy symbol $*$ to B . In both cases let B^* denote the resulting set of size a . Form the auxiliary $K_{a,a}$ between the gaps $g_i = (a_i, a_{i+1})$ and B^* with labels

$$\lambda(g_i v) = h(a_i v) + h(a_{i+1} v) \quad (v \in B), \quad \lambda(g_i *) = h(a_i a_{i+1}) \quad (N \text{ odd}),$$

all of norm at most $2L$. A perfect matching of the auxiliary graph inserts each $v \in B$ into its gap, i.e. replaces the pair (a_i, a_{i+1}) by the path $a_i v a_{i+1}$, and, when N is odd, closes the dummy's gap by the direct edge $a_i a_{i+1}$. The result traverses $a_1, \dots, a_a$ in order with each vertex of B inserted exactly once, so it is a Hamilton cycle H of K_N , and $h(H)$ equals the auxiliary label sum. Let μ be the mean auxiliary label sum over perfect matchings, i.e. $\frac{1}{a}\lambda(K_{a,a})$. Each edge of $E(A, B)$ occurs in two labels and, for odd N , each edge $a_i a_{i+1}$ in one, so with $Y = \sum_{i=1}^a h(a_i a_{i+1})$,

$$\mu = \frac{2}{a}T \quad (N \text{ even}), \quad \mu = \frac{2}{a}T + \frac{1}{a}Y \quad (N \text{ odd}).$$

If N is even, $a = \frac{N}{2}$ and $p = \frac{N}{2(N-1)}$, so $\frac{2p}{a} = \frac{2}{N-1}$ and $\mu - \frac{2}{N-1}S = \frac{2}{a}(T - pS)$. If N is odd, $a = \frac{N+1}{2}$ and $p = \frac{N+1}{2N}$, so $\frac{2p}{a} = \frac{2}{N}$ and, using $\frac{2}{N} - \frac{2}{N-1} = -\frac{2}{N(N-1)} = -1/\binom{N}{2}$,

$$\mu - \frac{2}{N-1}S = \frac{2}{a}(T - pS) + \frac{Y}{a} - \frac{S}{\binom{N}{2}}.$$

Since Y is a sum of a labels and S a sum of $\binom{N}{2}$ labels, $\|Y/a\| \leq L$ and $\|S/\binom{N}{2}\| \leq L$; and $\|\frac{2}{a}(T - pS)\| \leq \frac{2}{a} \cdot 2LdN \leq 8Ld$ because $a \geq \frac{N}{2}$. Hence in both cases

$$\left\| \mu - \frac{2}{N-1}S \right\| \leq 8Ld + 2L.$$

Proposition 2.4 applied to the auxiliary $K_{a,a}$ with label bound $2L$ gives a perfect matching whose label sum is within $6LdB_d$ of μ ; the corresponding Hamilton cycle H satisfies $\|h(H) - \frac{2}{N-1}h(K_N)\| \leq 6LdB_d + 8Ld + 2L$.

The centring vectors. A uniformly random Hamilton cycle of $K_{n,n}$ or K_N contains each edge with probability $e(H)/e(G)$ by edge-transitivity, so its expected label sum is $\frac{e(H)}{e(G)}h(G)$, which equals $\frac{2}{n}h(K_{n,n})$, respectively $\frac{N}{2}h(K_N) = \frac{2}{N-1}h(K_N)$. $\square$

Corollary 2.3 (Recolouring form). *Let $k \geq 2$ and let c be a colour-balanced k -edge-colouring of the host graph. If the host is $K_{n,n}$ with $n \geq 2$ and $k \mid 2n$, some Hamilton cycle can be made colour-balanced by recolouring at most $3k(\lceil \log_2 k \rceil + 2)$ of its edges. If the host is K_N with $N \geq 3$ and $k \mid N$, some Hamilton cycle can be made colour-balanced by recolouring at most $3k(\lceil \log_2 k \rceil + 2) + 4k + 1$ of its edges.*

Proof. Give every edge of colour j the label $h(e) = e_j \in (\mathbb{R}^k, \|\cdot\|_1)$, so $\|h(e)\|_1 = 1$ and $h(F) = (m_1(F), \dots, m_k(F))$ is the vector of colour counts of F . If the host $K_{n,n}$ is colour-balanced, each colour class has n^2/k edges and $\frac{2}{n}h(K_{n,n}) = \frac{2n}{k}\mathbf{1}$; if K_N is colour-balanced, $\frac{2}{N-1}h(K_N) = \frac{2}{N-1} \cdot \frac{1}{k} \binom{N}{2} \mathbf{1} = \frac{N}{k} \mathbf{1}$. A Hamilton cycle has $2n$, respectively N , edges, and $q = 2n/k$, respectively $q = N/k$, is an integer by hypothesis. If a subgraph F has kq edges and colour counts $m_1, \dots, m_k$, the least number of its edges that must be recoloured to make it colour-balanced is

$$\rho(F) = \sum_{j=1}^k (m_j - q)_+ = \frac{1}{2} \sum_{j=1}^k |m_j - q| = \frac{1}{2} \|h(F) - q\mathbf{1}\|_1 :$$

every surplus edge of an over-represented colour must be recoloured, and recolouring exactly the surplus edges into deficient colours (possible since $\sum_j (m_j - q)_+ = \sum_j (q - m_j)_+$ as $\sum_j m_j = kq$) achieves this. **Theorem 2.1** with $d = k$, $L = 1$ gives a Hamilton cycle H with $\|h(H) - q\mathbf{1}\|_1 \leq 6kB_k$, respectively $\leq 6kB_k + 8k + 2$; halving yields $\rho(H) \leq 3kB_k$, respectively $3kB_k + 4k + 1$. $\square$

B Report of the LLM referee (verbatim)

The following is the unedited report of the adversarial referee agent (claude-opus-5, 2026-09-17) on the *original* writeup. Its step numbers and lemma numbers refer to that writeup, not to this note (writeup Lemma 3, 4, 5, Proposition 6, Lemma 7 are [Lemmas 3.1 to 3.4](#) and [Proposition 2.4](#) here; its $f_{\tilde{f}}$ is the paper's f_h). The scripts it mentions are in `verification_astra/scripts/2604.09449__03/`. Treat it as machine output, not as an authority.

B.1 Interpretation

The interpretation is exactly the intended one. I downloaded the TeX source of arXiv:2604.09449v2 (the current version, posted 2026-07-11) and checked the problem statement and every definition the writeup implicitly relies on.

- The paper defines, for a subgraph $G' \subseteq G$ and $\tilde{f}: E(G) \rightarrow \mathbb{R}^k$,

$$f_{\tilde{f}}(G') = \left\| \tilde{f}(G') - \frac{e(G')}{e(G)} \tilde{f}(G) \right\|_1,$$

under the hypothesis $\|\tilde{f}(e)\|_1 \leq 1$. (Confirmed independently in the sibling report `verification/2604.09449__00.md`.)

- Problem 6.4 reads verbatim: “Improve the bounds on $f_{\tilde{f}}(H)$ for a Hamilton cycle H in complete or complete bipartite graphs.” The surrounding paragraph fixes the baseline: $\Delta = 2$ in `thm:bounded_deg_bound` ($f_{\tilde{f}} = O(\Delta^2 k^2 \sqrt{\log 2\Delta})$) gives $O(k^2)$, and “our methods imply that this bound also holds for Hamilton cycles in complete bipartite graphs”.
- The writeup’s centring vectors are $\frac{2}{n} \tilde{f}(K_{n,n})$ and $\frac{2}{N-1} \tilde{f}(K_N)$. These are *exactly* $\frac{e(H)}{e(G)} \tilde{f}(G)$: $2n/n^2 = 2/n$ and $N/\binom{N}{2} = 2/(N-1)$. So Theorem 1, specialised to $X = (\mathbb{R}^k, \|\cdot\|_1)$, $d = k$, $L = 1$, is literally a bound on the paper’s $f_{\tilde{f}}(H)$, with no reinterpretation and no weakening.
- The writeup is *more* general than the problem asks (arbitrary d -dimensional norm, not just ℓ_1), and covers both hosts requested. No sub-class restriction: the divisibility hypotheses ($k \mid 2n$, $k \mid N$) appear only in the *recolouring* Corollary 2, where they are necessary for “colour-balanced” to be achievable at all, and are stated.

A reasonable author of Problem 6.4 would consider it answered (an improvement was asked for, not the optimal order; the writeup is explicit that it does not determine the order).

Understated claim. The verdict line sells the result short. The substantive ingredient, Proposition 6, bounds $f_{\bar{f}}(M)$ for perfect matchings of $K_{r,r}$ by $3kB_k = O(k \log k)$. That is a direct improvement of the paper's $\text{thm:vec-bipartite}(O(k^2))$, which the authors call “a central tool, which we use throughout the paper” and about which they write: “any improvement to the general k^2 bound in thm:vec-bipartite immediately gives an equal improvement to the k^2 factor in the bounds of each of our other results.” If correct, the writeup therefore also improves Theorems 1.1, 1.2, 1.4, 1.5, 1.6 of the source paper. This raises the stakes and is the main reason I audited every line rather than spot-checking.

B.2 Step-by-step findings

step	label	note
1. $B_d = \lceil \log_2 d \rceil + 2$; Theorem 1 statement	VALID	Well-posed. Centring is the expected label sums of uniformly random Hamilton cycles $(2n/n^2, N/\binom{N}{2})$ and coincide with the paper's $f_{\bar{f}}$ centring.
2. Corollary 2 (recolouring form)	VALID	Arithmetic checked: $\frac{1}{2} \cdot 6LdB_d = 3kB_k$ and $\frac{1}{2}(6kB_k + 8k + 2) = 3kB_k + 4k + 1$.
3. Lemma 3 (necklace halving with $\leq d$ cuts)	VALID	This is the Hobby–Rice / 2-thief necklace theorem; the Borsuk–Ulam proof given is the textbook one. F is odd (lengths rx_i^2 are invariant under $x \mapsto -x$, signs flip) and continuous (a coordinate $\rightarrow 0$ has interval length $\rightarrow 0$; $\text{sgn}(0) = 0$ is harmless since that interval is null). $d + 1$ intervals give $\leq d$ internal cuts. Numerically confirmed for $d = 2$.
4a. σ via $Q(v) = P(\sigma(v))$, concatenated cyclic orders	VALID	Standard; with this order $\sigma(v_i) = v_{i+1}$ inside each block.
4b. Eq. (7) halving identity	VALID	Re-derived: $\sum_i (s_i - \frac{1}{2})(h(P_i) - h(Q_i))$ minus the vanishing $\sum_i (\alpha_i - \frac{1}{2})(\cdot)$ gives $\sum_i (s_i - \alpha_i)(\cdot)$.
4c. Eq. (8) $\sum_i s_i - \alpha_i \leq d/2$	VALID	If $\alpha_i \in (0, 1)$ then $\mathbf{1}_A$ is non-constant on I_i , so I_i 's interior contains a cut point; interiors are disjoint, so at most d such i , each contributing $\leq 1/2$.
4d. Eq. (9) $\ z - \text{mid}\ \leq Ld$	VALID	$2L \cdot d/2$. Note z is only a “one edge per V -vertex” selection at this stage, correctly flagged.
4e. $\leq d$ changes in the linear sequence $s_1, \dots, s_r$	VALID	The subtle point, and it is right: s_i is attained by $\mathbf{1}_A$ at some non-cut interior point t_i of I_i (for $\alpha_i \in (0, 1)$ both values are attained on positive measure, so any tie-break works); a change between t_i, t_{i+1} needs a cut in the disjoint interval (t_i, t_{i+1}) .
4f. Eq. (10) $b \leq 2d$ cyclic changes	VALID	Per block, cyclic $= c + [\text{first} \neq \text{last}] \leq 2c$ (the bracket forces $c \geq 1$); $\sum_{\text{blocks}} c \leq d$ since block-boundary changes are not internal to any block.
4g. degree $= 1 + s_i - s_{i-1}$; $b/2$ vertices of degree 2 and $b/2$ of degree 0	VALID	$P(v_i)$ is hit by v_i iff $s_i = 1$ and by v_{i-1} iff $s_{i-1} = 0$. In a cyclic binary word, $\#(01) = \#(10)$.
4h. repair; Eq. (11) $\leq 2Ld$	VALID	The $b/2$ deleted edges have distinct V -endpoints (each V -vertex carries one selected edge), so exactly $b/2$ vertices are freed and can be matched to the $b/2$ degree-0 U -vertices (host complete). Cost $2L \cdot b/2 = Lb \leq 2Ld$. Total $3Ld$, independent of r — the crux, and it is genuinely r-free.
5a. Lemma 5, error recursion	VALID	Re-derived: $\ z_v - x_v\ \leq D[\text{het}] + \frac{1}{2}(\ z_{v_1} - x_{v_1}\ + \ z_{v_2} - x_{v_2}\)$, unrolling to $D \sum_j 2^{-j} H_j$; leaves have zero error.
5b. $H_j \leq \min(2^j, m - 1)$	VALID	Heterogeneous nodes at a fixed depth have disjoint leaf intervals, each containing a distinct one of the $m - 1$ block boundaries.

step	label	note
5c. $\sum_{j \geq 0} \min(1, (m-1)2^{-j}) \leq \lceil \log_2(m-1) \rceil + 2$	VALID	Re-derived: $t + \sum_{j \geq t} (m-1)2^{-j} \leq t + 2$ with $t = \lceil \log_2(m-1) \rceil$ and $2^t \geq m-1$. With Carathéodory $m \leq d+1$, this is $\leq B_d$. $m=1$ handled; $d=1$ gives $B_1=2$, consistent.
5d. dyadic $\rightarrow$ arbitrary coefficients	VALID (terse)	Correct: S finite, so a constant subsequence of root representatives exists and the bound passes to the limit. Dropping a coefficient to 0 only decreases m .
6. Proposition 6: $\ h(M) - \frac{1}{r}h(K_{r,r})\ \leq 3LdB_d$	VALID	Lemma 4 supplies the hypothesis of Lemma 5 for <i>all</i> pairs with $D = 3Ld$; a uniform random perfect matching uses each edge with probability $(r-1)!/r! = 1/r$, so the centre lies in $\text{conv}(S)$.
7. §5, bipartite Hamilton reduction (17)–(19)	VALID	Each original edge $a_i b$ occurs in exactly the two labels $\lambda(g_{i-1}b), \lambda(g_i b)$, so $\sum_{i,b} \lambda = 2h(K_{n,n})$ and the mean is $\frac{2}{n}h(K_{n,n})$. Labels have norm $\leq 2L$, so Prop. 6 yields $3(2L)dB_d = 6LdB_d$. $n=2$ works.
8a. Lemma 7, $\mathbb{E}I_e = p = \frac{2ab}{N(N-1)}$ and $\mathbb{P}(I_e = I_f = 1) = p/2$ for adjacent e, f	VALID	Verified exactly (Fraction arithmetic, all $\binom{N}{a}$ partitions) for $N=3, \dots, 10$.
8b. Eq. (21) $ \text{Cov} = p(\frac{1}{2} - p) \leq 1/N$	VALID	$ \frac{1}{2} - p = \frac{1}{2(N-1)}$ (N even), $\frac{1}{2N}$ (N odd); $p \leq 1$, and $\frac{1}{2(N-1)} \leq \frac{1}{N}$ for $N \geq 2$. Verified exactly for $N \leq 10$.
8c. Eq. (22) disjoint-pair covariance and the $4a(a-1)b(b-1)/(N)_4$ joint probability	VALID	Both formulas verified exactly for $N=4, \dots, 10$; and $\leq 4/N^2$ holds for all $N \geq 4$ (tightest at $N=4$: $2/9 \leq 1/4$).
8d. Eq. (23) $\text{Var } Z \leq \frac{17}{8}L^2N^2 < 4L^2N^2$	VALID	Pair counts $\binom{N}{2}, 3\binom{N}{3}, 3\binom{N}{4}$ correct; the three terms are $\leq N^2/8 + N^2 + N^2$. Exact enumeration of $\sum_{e,f} \text{Cov} $ for $N=4..9$ stays well under the writeup's own expression.
8e. Auerbach basis, $\ x\ \leq \sum_j \varphi_j(x) $, $\mathbb{E}\ T\ \leq 2LdN$	VALID	The determinant-maximisation construction of an Auerbach basis is standard and correctly described; $\mathbb{E}[\varphi_j(T)] = 0$, so $\mathbb{E} \varphi_j(T) \leq \sqrt{\text{Var}} \leq 2LN$. $\frac{1}{a}(2T)$ resp. $\frac{1}{a}(2T+Y)$.
9a. Eq. (27) mean auxiliary matching sum	VALID	Re-derived symbolically; both identities are exact.
9b. Eqs. (28),(29): $2p/a = \frac{2}{N-1}$ (N even), $= \frac{2}{N}$ (N odd) and $\frac{2}{N} - \frac{2}{N-1} = -1/\binom{N}{2}$	VALID	$(2/a) \cdot 2LdN \leq 8Ld$ using $a \geq N/2$; $\ Y/a\ \leq L, \ S/\binom{N}{2}\ \leq L$.
9c. Eq. (30) $\leq 8Ld + 2L$	VALID	$N=3$: the only Hamilton cycle is K_3 and $\frac{2}{N-1}S = S$, so the deviation is 0.
9d. final bound (2); $N=3$ base case	VALID	$\rho(F) = \sum_j (m_j - q)_+ = \frac{1}{2} \sum_j m_j - q $ since $\sum_j m_j = kq$; the min-recolouring characterisation is correct. Colour-balanced hosts give $\frac{2}{n}h(K_{n,n}) = \frac{2n}{k} \mathbf{1}$ and $\frac{2}{N-1}h(K_N) = \frac{N}{k} \mathbf{1}$.
10. §7 translation, Eq. (31)	VALID	Verified by exhaustive code for $N=3..10$ (all $a!$ matchings, both parities) and for $K_{n,n}$.
11. Structural claim that the auxiliary matchings really give Hamilton cycles	VALID	

No step was labelled GAP or ERROR.

Relation to the earlier attempt in the prompt. The new writeup does not merely inherit: it *sharpens* Lemma 4 from $8Ld$ to $3Ld$ and, in doing so, replaces the earlier attempt's weakest passage — the earlier text chose s_i arbitrarily on “exceptional” intervals and then had to argue “rounding an exceptional position creates at most two further linear changes”, reaching $b \leq 2(d+2F) \leq 6d$ by a route that is hand-wavy about where the changes can occur. The new argument (realise s_i as $\mathbf{1}_A$ at a non-cut interior point) makes the $\leq d$ linear changes immediate and is correct. It also generalises Lemma 6/7 from the ℓ_1 colour setting to an arbitrary norm via an Auerbach basis. The three specific failure modes flagged for a “beats-the-requested-bound” claim are all absent: no sub-class restriction, the logarithm comes from an explicit dyadic tree whose level counts are proved (not from an assumed balanced split), and the recursion's base constant $3Ld$ is carried openly into the final $3LdB_d$.

B.3 Reference check

Everything the writeup uses externally is either proved in place or is a standard named theorem used correctly:

- **Borsuk–Ulam** (odd continuous $S^d \rightarrow \mathbb{R}^d$ has a zero) — used correctly in Lemma 3. Lemma 3 itself is the Hobby–Rice theorem / 2-thief necklace splitting; the writeup proves it rather than citing it, so there is nothing to misquote.
- **Carathéodory** ($x \in \text{conv}(S)$ in dimension d is a convex combination of $\leq d + 1$ points) — used correctly in Lemma 5.
- **Auerbach basis** — the determinant-maximisation existence proof sketched is the standard one.
- **Source paper**, arXiv:2604.09449v2 (TeX source downloaded and read):

- $f_{\vec{f}}(\vec{f}(G')) = \frac{f(G')}{f(G)} \frac{f(G)}{f(G')} \frac{f(G)}{f(G')}$ — confirmed, matches the writeup’s centring exactly.
- thm:vec-bipartite : $f_{\vec{f}}(M) = O(k^2)$ for perfect matchings of $K_{n,n}$ — confirmed; this is what Proposition 6 improves.
- $\text{thm:bounded_deg_bound}$: $O(\Delta^2 k^2 \sqrt{\log 2\Delta})$, giving $O(k^2)$ at $\Delta = 2$ — confirmed as the baseline for Hamilton cycles.
- Problem 6.4 — confirmed verbatim, still open in v2.
- Lower bounds thm:lower-Knn ($f_c \geq \sqrt{k/2}$ for colourings of $K_{k,t,k,t}$) and thm:lower_Kn — confirmed; **no contradiction** with an $O(k \log k)$ upper bound. In the general *vector* setting the sibling target 2604.09449__00 exhibits labellings with $f_{\vec{f}}(M) = k$ for every matching; I re-ran that construction and confirmed $k \leq 3kB_k$ for $k = 2, 4, 6, 8$, i.e. the new bound is consistent with, and within a log factor of, the best known lower bound in its own setting.

references_ok: **true**. The writeup invokes no result it misstates, and its one honest omission (not checking the source paper’s later revision) I have now discharged: v2 is the current version and still poses the problem with the $O(k^2)$ bound.

B.3.1 Priority check (step 2b) and self-contamination (step 2c)

- Semantic Scholar: arXiv:2604.09449 has **0 citations**.
- arXiv API abstract search for "colour-balanced" / "color-balanced" (all years, newest first) returns 5 papers; the only 2026 ones are the source paper itself and 2608.12013 *Layer barriers for colour-biased tight Hamilton cycles* (unrelated: colour-bias in hypergraphs, not representative subgraphs).
- No arXiv paper found with a **representative perfect matching** bound of $O(k \log k)$; searches over the Pardey–Rautenbach / Hollom / Banerjee–Hollom line turn up only the known chain $3k\sqrt{kt \log 2k}$ (2105.05661) $\rightarrow 4^{k^2}$ (2410.07993) $\rightarrow O(k^2)$ (2604.09449). arXiv:2601.18789 (Banerjee–Hollom, *Uniformly balanced H-factors*) is the predecessor the source paper supersedes, not an improvement of it.
- v2 of the source paper (2026-07-11) is the latest version, and its §6 still states Problem 6.4 and the $O(k^2)$ bound. No v3.
- I also looked for the general lemma as a named result (an approximately midpoint-closed set is within $D(\log_2 d + 2)$ of its convex hull; linear discrepancy of the bipartite perfect-matching polytope under d linear functionals). I found nothing matching. This is the one place where I cannot *prove* novelty: the discrepancy-theory literature (Beck–Fiala, Steinitz/Bárány–Grinberg, Banaszczyk-type vector balancing) is large, and an $O(d)$ or $O(d \log d)$ rounding bound for the Birkhoff polytope with d extra linear constraints could plausibly exist as folklore. I checked that the two obvious routes do **not** give it: a vertex-of-LP argument fails because the extra constraints involve every edge (so no constraint can be dropped cheaply), and the classical $\{0, 1\}$ -rounding lemma ($\|\sum(\lambda_i - \varepsilon_i)v_i\| \leq d$) does not respect the matching structure. I record this as a novelty caveat, not as prior art.
- **Self-contamination**: a web search for this claim against the paper surfaces github.com/graph-theory-AI/Graph-Theory-LLM-Proofs PR #3 — this campaign’s own pull request. That is *not* prior art and I have not treated it as such. No independent paper, author, date or venue was found for the $O(k \log k)$ bound.

Conclusion: **not ALREADY_KNOWN** on the evidence available.

B.4 Computational check

Scripts in `verification_astra/scripts/2604.09449__03/` (python3, numpy + exact Fraction).

`cov_check.py` — **Lemma 7’s probabilities, exact, all $\binom{N}{a}$ partitions**, $N = 3..10$. All of the writeup’s claims hold as *exact identities*:

N	p	$\mathbb{P}(I_e I_f = 1)$, adjacent	$p/2$	Cov, disjoint	claimed (22)	$\leq 4/N^2$
4	2/3	1/3	1/3 ✓	2/9	2/9 ✓	✓ ($0.222 \leq 0.25$)
5	3/5	3/10	3/10 ✓	1/25	1/25 ✓	✓
6	3/5	3/10	3/10 ✓	1/25	1/25 ✓	✓
7	4/7	2/7	2/7 ✓	4/245	4/245 ✓	✓
8	4/7	2/7	2/7 ✓	4/245	4/245 ✓	✓
9	5/9	5/18	5/18 ✓	5/567	5/567 ✓	✓
10	5/9	5/18	5/18 ✓	5/567	5/567 ✓	✓

The stated joint probability $4a(a-1)b(b-1)/(N(N-1)(N-2)(N-3))$ is exact for every $N \geq 4$ tested, and $|\text{Cov}(I_e, I_f)| \leq 1/N$ holds for adjacent pairs throughout.

`var_check.py` — **the variance bound (23), exact**. $\sum_{e,f} |\text{Cov}(I_e, I_f)|$ versus the writeup’s bounding expression $\frac{1}{4}\binom{N}{2} + \frac{2}{N}\binom{N}{3} + \frac{8}{N^2}3\binom{N}{4}$: $N = 4$: 5.33 vs 9.00; $N = 5$: 7.20 vs 19.29; $N = 6$: 14.40 vs 33.75; $N = 7$: 17.14 vs 52.39; $N = 8$: 27.43 vs 75.25; $N = 9$: 31.11 vs 102.33. All within, and the expression is below $\frac{17}{8}N^2$ and $4N^2$ in every case.

`brute.py` — **exhaustive existence checks** (ratio = best attainable value ÷ claimed bound; must be ≤ 1):

- Lemma 4 (bound $3Ld$), random ℓ_1 -normalised labels, 60 instances each: $(r, d) = (5, 2)$ 0.140, $(5, 3)$ 0.113, $(6, 2)$ 0.074, $(6, 3)$ 0.095, $(6, 4)$ 0.088, $(7, 3)$ 0.048.
- Proposition 6 (bound $3LdB_d$): ratios 0.007–0.019 over the same instances.
- Theorem 1(1), exhaustive over Hamilton cycles of $K_{n,n}$: $(n, d) = (4, 2)$ 0.034, $(5, 2)$ 0.015, $(5, 3)$ 0.012, $(6, 3)$ 0.006.
- Theorem 1(2), exhaustive over **all** Hamilton cycles of K_N : $(N, d) = (5, 2)$ 0.019, $(6, 2)$ 0.014, $(6, 3)$ 0.007, $(7, 3)$ 0.004.

- Structural: for $N = 3..10$, every one of the $a!$ auxiliary matchings (dummy included for odd N) yields a genuine Hamilton cycle of K_N — **True** in all cases.
- The adversarial labelling of the sibling target ($f_{h(m)}=k$ for every matching) attains exactly k for $k = 2, 4, 6, 8$ against bounds $3kB_k = 18, 48, 90, 120$ — consistent, and it confirms the $\Omega(k)$ lower bound in this setting, so the claimed $O(k \log k)$ is near-optimal rather than implausibly strong.

adversarial.py — **hill-climbing search for a violating labelling** (maximising attained $\div$ bound over the label configuration, ℓ_1 -normalised labels, several restarts): Lemma 4 at $(r, d) = (4, 1)$ **0.333**, $(4, 2)$ 0.314; Proposition 6 at $(5, 1)$ 0.133, $(5, 2)$ 0.045, $(6, 1)$ 0.167, $(6, 3)$ 0.016. Nothing approaches 1. The same script confirms Lemma 3 numerically for $d = 2, r = 6$: over random $\mathbb{R}^2$ -valued step functions, a 2-cut halving set always exists, worst residual 0.020 at grid resolution 0.03.

No computational claim failed. computation_run: **true**.

B.5 Caveats

1. **Novelty, not correctness, is the residual risk.** I could not find the result or its load-bearing lemmas in the literature, and the source paper’s v2 still poses the problem — but Lemma 5 (approximate midpoint closure $\Rightarrow$ log-factor convexification) is a clean, general statement of the kind that could exist in discrepancy theory under other language. A human referee should search that literature before publication. The only web hit connecting this claim to the paper is this campaign’s own repository (self-contamination, discounted).
2. **The announced claim understates what is proved.** Proposition 6 improves the source paper’s `thm:vec-bipartite` from $O(k^2)$ to $O(k \log k)$; by the authors’ own remark this propagates to *all* of their main theorems. The writeup mentions this only in passing (“It also establishes the corresponding $O(k \log(k+1))$ bipartite perfect-matching estimate”).
3. **Order not determined.** The writeup is explicit that it does not rule out $O(k)$. In the general vector setting the truth is now pinned between k (sibling counterexample) and $3kB_k$; for *colourings* the paper’s lower bound is only $\sqrt{k/2}$ and its $O(\sqrt{k})$ conjecture is untouched.
4. **Restricted Hamilton-cycle family.** The complete-graph construction only ever produces cycles that alternate between A and B (plus one A -internal edge when N is odd). That is legitimate for an upper bound, but it means the bound is not tight by construction and the argument says nothing about which Hamilton cycles achieve it.
5. **Edge cases.** $n \geq 2$ (bipartite) and $N \geq 3$ (complete) are required and stated; Lemma 7 needs $N \geq 4$ (the $N-3$ denominator) and $N=3$ is handled separately; $d=1$ works with $B_1=2$. Disconnected/empty hosts do not arise. Corollary 2 additionally needs the host to be colour-balanced and $k \mid 2n$ resp. $k \mid N$ — necessary, and stated.
6. **Terse but repairable-as-written passages** (none rise to GAP): the dyadic-to-arbitrary limiting argument in Lemma 5; the “at most $4/N^2$ ” assertion in (22) (true for all $N \geq 4$, I checked the algebra and the small cases); the one-line Auerbach basis construction.
7. **Constants are not optimised** and the ℓ_1 /general-norm constants differ by the factor $\frac{1}{2}$ used in the recolouring translation; I checked that translation is applied consistently in (3) and (4).

B.6 Referee summary

I set out to break this writeup and could not. Reading it against the actual TeX of arXiv:2604.09449v2, the interpretation is exact — the writeup’s centring vectors are literally the paper’s $\frac{e(H)}{e(G)} \tilde{f}(G)$, the hypothesis $\|h(e)\| \leq L$ specialises to the paper’s $\|\tilde{f}(e)\|_1 \leq 1$, and Problem 6.4 asks precisely for an improvement over $O(k^2)$, which $3kB_k + 4k + 1 = O(k \log(k+1))$ delivers for both requested hosts and, in fact, for an arbitrary d -dimensional norm. Every one of the twenty-odd steps I isolated is valid: I re-derived the error recursion of the dyadic tree, the $\sum_j \min(1, (m-1)2^{-j}) \leq \lceil \log_2(m-1) \rceil + 2$ estimate, the cyclic-change and degree bookkeeping in the splicing lemma (the one genuinely delicate argument, and the place the earlier attempt was hand-wavy — this version fixes it by realising each rounded bit as $\mathbf{1}_A$ at a non-cut point, which makes “ $\leq d$ changes” immediate), and all of the K_N cut algebra. Exact rational computation confirms every probability and covariance in Lemma 7 for $N \leq 10$ including the delicate “ $\leq 4/N^2$ ” at $N=4$; exhaustive enumeration confirms Theorem 1 for $K_{n,n}$ up to $n=6$ and K_N up to $N=7$, the structural claim that the auxiliary matchings really give Hamilton cycles for $N \leq 10$, and adversarial hill-climbing over labellings never pushes any claimed bound past a third of its budget. The three failure modes typical of a “beats-the-asked-for-bound” claim are all absent. The one thing I cannot certify is novelty against the wider discrepancy literature: the search evidence is clean (0 citations, no follow-up on arXiv, v2 still poses the problem, and the only web hit is this campaign’s own repository, which I discount as self-contamination), but Lemma 5 is a general enough statement that a human expert should check it against Beck–Fiala/Steinitz-type folklore before submission. Worth emphasising for whoever acts on this: the result is *larger* than its verdict line says — Proposition 6 improves the source paper’s cornerstone $K_{n,n}$ matching theorem, which the authors state would improve every other bound in their paper by the same factor.

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